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A PHYLOGENY OF THE CHLOROPLAST GENE *RBC*L IN THE LEGUMINOSAE: TAXONOMIC CORRELATIONS AND INSIGHTS INTO THE EVOLUTION OF NODULATION[1]

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Phylogenetic analyses of the chloroplast-encoded *rbc*L gene in Leguminosae are consistent with previous hypotheses in suggesting that the family as a whole is monophyletic, but that only two of its three subfamilies are natural. The earliest dichotomies in the family appear to have involved tribes Cercideae or Cassieae (subtribe Dialiinae), followed by Detarieae/Macrolobieae, all of which are members of subfamily Caesalpinioideae. The remainder of the family is divided into two clades: (1) Mimosoideae and the caesalpinioid tribes Caeasalpinieae and Cassieae (subtribes Ceratoniinae and Cassiinae); (2) Papilionoideae. Basal groups within Papilionoideae are, as expected, elements of the grade tribes Sophoreae and Swartzieae. Major clades within Papilionoideae include: (1) a Genistoid Alliance comprising Genisteae, Crotalarieae, Podalyrieae, Thermopsideae, Euchresteae, and also some Sophoreae; (2) a clade marked by the absence of one copy of the chloroplast inverted repeat, with which are associated Robinieae, Loteae, and some Sophoreae; (3) Phaseoleae, Desmodieae, Psoraleeae, and most Millettieae, a group also marked by presence of pseudoracemose inflorescences; and (4) a well-supported clade comprising Aeschynomeneae, Adesmieae, and some Dalbergieae. Nodulation is most parsimoniously optimized on the *rbc*L strict consensus tree as three parallel gains, occurring in Papilionoideae, the caesalpinioid ancestors of Mimosoideae, and in the genus *Chamaecrista* (Caesalpinieae: Cassieae).

Key words: Leguminosae; nodulation; phylogeny; *rbc*L; taxonomy.

The Leguminosae are the third largest family of angiosperms, with three subfamilies comprising some 650 genera and $>$ 18 000 species (Polhill and Raven, 1981; Polhill, 1994). It has long been suspected that many of the higher taxonomic groupings in the family are unnatural. For example, precladistic interpretations of the family hypothesized polyphyly/paraphyly of subfamily Caesalpinioideae (Polhill, Raven, and Stirton, 1981), and of numerous tribes, including putatively basal groups of Caesalpinioideae (Caesalpinieae: Polhill and Vidal, 1981) and Papilionoideae (Sophoreae: Polhill, 1981f), as well as more derived elements such as the largest papilionoid tribe, Phaseoleae (Lackey, 1981).

In the past several years, considerable progress has been made in constructing objective phylogenetic hypotheses in the family, many of which are summarized in the most recent volume of the *Advances in Legume Systematics* series (Crisp and Doyle, 1995) and are reflected in the latest comprehensive taxonomic treatment of the family (Polhill, 1994). Several of these corroborate the earlier intuitive studies. Most of these phylogenetic studies have been confined to genera in a single tribe, or in some cases to a few closely related tribes, and many have utilized molecular characters, such as chloroplast DNA (cpDNA) restriction mapping (e.g., Doyle and Doyle, 1993), nuclear ribosomal DNA internal transcribed spacer (nrDNA ITS) sequences (e.g., Wojciechowski et al., 1993), or phytochrome genes (Mathews, Lavin, and Sharrock, 1995).

Two notable exceptions to these generalizations are the morphological cladistic analyses of Tucker and Douglas (1994) and Chappill (1995). The first of these sampled relatively few taxa, and included a number of floral ontogenetic characters. The Chappill analysis was comprehensive in its representation of the family's subfamilies and tribes, and included a few molecular characters. Both were acknowledged by their authors to be quite preliminary, and the relationships hypothesized from the two studies differed dramatically (see below for discussion). There is clearly a need, therefore, for additional extensive analyses of the Leguminosae. Sequences of the chloroplast gene ribulose-1-5-bisphosphate carboxylase/oxygenase (*rbc*L) have been used (with variable success) to produce phylogenetic hypotheses within angiosperm families (see, for example, various contributions in *Annals of the Missouri Botanical Garden*, vol. 80, 1993). In the legumes, the sole published analysis of this gene (Käss

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TABLE 1. Sources of Leguminosae *rbc*L sequences. The classification and tribal/generic ordering of Leguminosae of Polhill (1994) are used. Specimen information, generally in the form (voucher; herbarium abbreviation), is given after each species. GenBank accession numbers are U74183–U74259.

Subfamily Caesalpinioideae

Tribe Caesalpinieae: *Gleditsia* group: *Gymnocladus* L.: G. dioicus (L.) K. Koch (Lavin s.n.; BH). *Peltophorum* group: *Peltophorum* (Vogel) Benth.: *P. pterocarpum* (DC) Backer ex K. Heyne (PI 221228; USDA); "*P. peltatum*" (no voucher information). *Caesalpinia* group: *Caesalpinia* L.: *C. pulcherrima* (L.) Sw. (Doyle 1465; BH). *Dimorphandra* group: *Erythrophleum* G. Don: *E. ivorense* A. Chev. (McKey C91-11; K); *Sclerolobium* group: *Tachigali* Aubl. *T. paniculata* Aubl. (Vásquez & Jaramillo 12942; MO)

Tribe Cassieae: Subtribe Ceratoniinae: *Ceratonia* L.: *C. siliqua* L. (MBG 897520; MO). Subtribe Dialiinae: *Apuleia* Mart.: *A. leiocarpa* (Vogel) J. F. Macbr. (R. T. Pennington s.n. 7/24/91; K); *Dialium* L.: *D.* sp. (McKey C91-5; K). Subtribe Cassiinae: *Cassia* L.: *C. fistula* L. (Doyle 1472; BH); *Chamaecrista* Moench: *C. fasciculata* (Michx.) Greene (Doyle s.n.; BH); *Senna* Mill.: *S. alata* (L.) Roxb. (Doyle 1473; BH).

Tribe Cercideae (5/6): Subtribe Cercidinae: *Cercis* L.: *C. canadensis* L. (Doyle 1527; BH). Subtribe Bauhiniinae: *Bauhinia* L.: *B.* sp. (cult. Michigan State U.).

Tribe Detarieae: *Brownea* group: *Brownea* Jacq.: *B.* sp. (Doyle 1538; BH)

Tribe Macrolobieae: *Macrolobium* group: *Macrolobium* Schreb.: *M. acaciifolium* (Benth.) Benth. (Henderson 693; NY).

Subfamily Mimosoideae

Tribe Parkieae: *Parkia* R. Br.: *P. roxburghii* G. Don (Luckow s.n.; BH).

Subfamily Papilionoideae

Tribe Swartzieae: *Aldina* group: *Aldina* Endl.: *A. latifolia* Benth. (Henderson 699; NY); *Cordyla* Lour.: *C. africana* Lour. (Polhill & Richardson 5312; K). *Ateleia* group: *Ateleia* (DC.) Benth.: *A. herbert-smithii* Pittier (T. D. Pennington s.n.; K). *Lecointea* group: *Holocalyx* Micheli: *H. balansae* Micheli (R. T. Pennington s.n., K); *Swartzia* group: *Swartzia* Schreb.: *S.* sp. (Henderson 695; NY).

Tribe Sophoreae: *Myroxylon* group: *Acosmium* Schott: *A. dasycarpum* (Vogel) Yakovlev (S. Bridgewater & T. R. Pennington 5235; K); *Luetzelburgia* Harms: *L.* sp. (R. T. Pennington s.n. 7/24/91; K); *Myrospermum* Jacq.: *M. sousanum* A. Delgado & M. C. Johnst. (Doyle 1524; BH); *Myroxylon* L. f.: *M. balsamum* (L.) Harms (Royal Botanic Gardens, Sydney, 13492). *Angylocalyx* group: *Angylocalyx* Taub.: *A. braunii* Harms (R. T. Pennington s.n.; K); *Castanospermum* A. Cunn. ex Hook.: *C. australe* A. Cunn. & C. Fraser (Uhl 600; BH); *Xanthocercis* Baill.: *X. zambesiaca* (Baker) Dumaz-le-Grand (RBG Kew living collection, 354 7203405). *Ormosia* group: *Ormosia* Jacks.: *O. emarginata* (Hook, & Arn.) Benth. (T. Kajita 95051602; TUS); *Pericopsis* Thwaites: *P. mooniana* (Thwaites) Thwaites (Pacific Tropical Garden, Kauaii, Hawaii, 74s535). *Baphia* group: *Baphia* Afzel. ex Lodd: *B. massaiensis* Taub. (Zimbabwe Botanic Garden 129). *Dussia* group: *Dussia* Krug & Urb. ex Taub.: *D. tessmannii* Harms (T. D. Pennington 13575; K). *Sophora* group: *Cadia* Forssk.: *C. purpurea* (G. Piccioli) Aiton (Royal Botanic Gardens, Sydney, 2434); *Bolusanthus* Harms: *B. speciosus* (Bolus) Harms (Doyle 1479; BH); *Cladrastis* Raf.: *C. sikokiana* (Makino) Makino (T. Kajita 96083002; TUS); *Maackia* Rupr.: *M. floribunda* (Miq.) Takeda (H. Hoshi 279; TUS); *Salweenia* Baker f. *S. wardii* Baker f. (Yuan s.n.; BH); *Sophora* L.: *S. bhutanica* H. Ohashi (H. Hara et al. 13635; TUS); *S. japonica* L. = *Styphnolobium japonicum* (L.) Schott (T. Kajita 96083001; TUS).

Tribe Dipterygeae: *Dipteryx* Schreb.: *D. odorata* (Aubl.) Willd. (R. T. Pennington s.n. 7/24/91; K)

Tribe Dalbergieae: *Andira* Juss.: *A. inermis* (W. Wright) DC. (T. D. Pennington 13345; K); *Dalbergia* L. f.: *D. hupeana* Hance (T. Kajita 96083003; TUS); *Machaerium* Pers.: *M. lunatum* (L. f.) Ducke (T. Kajita & T. Watanabe 94112604; TUS)

Tribe Abreae: *Abrus* L.: *A. precatorius* L. (Doyle 1291; BH)

Tribe Millettieae: *Derris* Lour.: *D. laxiflora* Benth. (H. Ohashi et al. 14006; TUS); *Austrosteenisia* Geesink = *Kunstleria* Prain: *A. blackii* (F. Muell.) Geesink (= *K. blackii* (F. Muell.) Polhill) (Pedley 5005; K); *Millettia* Wight & Arn.: *M. japonica* (Siebold & Zucc.) A. Gray (T. Kajita 95051601; TUS); *Tephrosia* Pers.: *T. heckmanniana* Harms (Lavin 6197; BH); *Wisteria* Nutt.: *W.* sp. (Lavin 6205; BH)

Tribe Robinieae: *Robinia* L.: *R. pseudoacacia* L. (Doyle 1529; BH)

Tribe Indigofereae: *Indigofera* L.: *I. australis* Willd. (Doyle 1539; BH); *Phylloxylon* Baill.: *P. perrieri* Drake (Schrire et al. 2527; BH)

Tribe Desmodieae: Subtribe Desmodiinae: *Desmodium* Desv.: *D. podocarpum* DC., ssp. *oxyphyllum* (DC.) H. Ohashi (Y. Tateishi & T. Nemoto 20484; TUS). Subtribe Lespedizinae: *Campylotropis* Bunge: *C. griffithii* Schindl. (T. Nemoto 5889; TUS); *Kummerowia* Schindl.: *K. stipulacea* (Maxim.) Makino (T. Nemoto 8584; TUS); *Lespedeza* Michx.: *L. cuneata* (Dum. Cours.) G. Don. (J. Murata & T. Nemoto s.n.; TUS)

Tribe Phaseoleae: Subtribe Diocleinae: *Canavalia* DC.: *C. ensiformis* (L.) DC.: (Doyle 1107; BH). Subtribe Glycininae: *Glycine* Willd.: *G. tabacina* (Labill.) Benth. (Doyle 934; BH). Subtribe Kennediinae: *Hardenbergia* Benth.: *H. violacea* (Schneev.) Stearn (Doyle 1015; BH). Subtribe Clitoriinae: *Clitoria* L.: *C. ternatea* L. (Doyle 1507; BH).

Tribe Psoraleeae: *Bituminaria* Fabricius: *B. bituminosa* (L.) C. H. Stirt. (NF Weeden s.n.; BH); *Otholobium* C. H. Stirt.: *O. sericeum* (Poir.) C. H. Stirt. (RBG Kew living collection, 1986-2127; K)

Tribe Amorpheae: *Amorpha* L.: *A. fruticosa* L. (Doyle 1525; BH)

Tribe Aeschynomeneae: Subtribe Poiretiinae: *Zornia* J. F. Gmel.: *Z. cantoniensis* Mohlenbr. (Y. Tateishi, T. Nemoto, & T. Kajita 911116; TUS). Subtribe Stylosanthinae: *Arachis* L.: *A. hypogaea* L. (T. Kajita et al. 95070201; TUS)

Tribe Adesmieae: *Adesmia* DC.: *A. exilis* Clos (Doyle 1466; BH)

Tribe Hedysareae: *Hedysarum* L.: *H. vicioides* Turcz. (T. Kajita et al. 95060302; TUS)

Tribe Loteae: *Lotus* L.: *L. corniculatus* L. (Doyle 1530; BH); *Coronilla* L.: *C. varia* L. = *Securigera varia* (L.) Lassen (Doyle 1559; BH)

Tribe Brongniartieae: *Brongniartia* Kunth: *B. pacifica* McVaugh (Dorado 60; RSA)

Tribe Mirbelieae: *Chorizema* Labill.: *C. cordatum* Lindl. (cult. U.C. Riverside Bot. Gard.)

Tribe Bossiaeeae: *Goodia* Salisb.: *G. lotifolia* Salisb. (Lavin 6222; BH)

Tribe Podalyrieae: *Calpurnia* E. Mey.: *C. aurea* (Aiton) Benth. (Lavin 6198; BH); *Podalyria* Willd.: *P. calyptrata* (Retz.) Willd. (Doyle 1515; BH)

Tribe Crotalarieae: *Dichilus* DC.: *D. lebeckioides* DC. (McMurtry 6376; K)

Tribe Euchresteae: *Euchresta* Benn.: *E. horsfieldii* Benn. (T. Kajita 89072601; TUS)

and Wink, 1995) focused primarily on the papilionoid tribe Genisteae, but included outgroups representing other papilionoid tribes and a representative of the subfamily Caesalpinioideae. We report here on a comprehensive phylogenetic study of legume *rbc*L sequences, the results of which provide insights into the subfamilial and tribal taxonomy of the family, as well as into the evolution of nodulation.

MATERIALS AND METHODS

Total genomic DNA was isolated from individual plants of genera representing all three subfamilies of the Leguminosae (Table 1), pri-

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marily using the method of Doyle and Doyle (1987). Fresh material was used where available. Emphasis was on subfamilies Caesalpinioideae, where all five tribes were represented, and Papilionoideae, where taxa from 26 of the 30 tribes recognized by Polhill (1994) were included. Papilionoideae tribes not included in the main analysis were the small South African tribe Liparieae (five genera) and three tribes from the group united by the absence of the cpDNA inverted repeat (IR; Lavin, Doyle, and Palmer, 1990: Cicereae, Galegeae, Carmichaelieae), though one of these (Galegeae) was represented in some analyses. In addition to *rbc*L sequences listed in Table 1, several Papilionoideae sequences were obtained from the literature or from other laboratories: *Medicago sativa* (Trifolieae; Aldrich et al., 1986; GenBank X04975); *Pisum sativum* (Vicieae; Zurawski, Whitfield, and Bottomley, 1986; GenBank X03853); *Vigna unguiculata* (Phaseoleae subtribe Phaseolinae; J. D. Palmer, Indiana University, unpublished data); *Lupinus polyphyllus*, *Genista tinctoria*, *Cytisus scoparius*. (Genisteae; Käss and Wink, 1995; GenBank Z70052, Z70099, Z70086); *Anagyris foetida* (Thermopsideae; Käss and Wink, 1995; GenBank Z70122). Subfamily Mimosoideae, which is widely held to be monophyletic (e.g., Polhill, 1994), was represented only by the complete sequence of *Parkia* (Parkieae) and, in some analyses, by (1) the partial sequence of *Albizia julibrissin* (Mimoseae; nucleotides 511–1119), provided by G. Learn and M. T. Clegg (University of California, Riverside), which was used in the comprehensive analysis of flowering plant *rbc*L sequences of Chase et al. (1993), and (2) unpublished partial sequences (nucleotides 1115–1428) from a second Mimoseae (*Leucaena*), and from representatives of two of the three remaining mimosoid tribes, Acacieae (*Acacia*) and Ingeae (*Calliandra*), which were obtained from S. Maumont (Université de Toulouse). Within Caesalpinioideae, efforts concentrated on the putatively basal, paraphyletic tribe Caesalpinieae, where representatives of five of the nine informal generic groups recognized by Polhill (1994) were sampled. Similarly, the putatively basal tribes of Papilionoideae, Swartzieae and Sophoreae, were emphasized in sampling; in the latter tribe, 18 genera were included, representing all six of the informal generic groups of Polhill (1994; seven of the eight recognized by Polhill, 1981f).

Outgroup *rbc*L sequences available in GenBank were chosen for phylogenetic analyses to include taxa hypothesized to be related to Leguminosae based on either: (1) morphological data and taxonomic classification (e.g., Sapindaceae/Aceraceae, Connaraceae, Chrysobalanaceae, Krameriaceae, Crossosomataceae: Dickison, 1981), or (2) results of several published cladistic analyses that included legume *rbc*L sequences (e.g., Surianaceae, Polygalaceae, *Quillaja*: Chase et al., 1993; Fernando et al., 1993; Morgan and Soltis, 1993; Soltis et al., 1995). Additional sequences from taxa somewhat more distantly related on the basis of *rbc*L studies (e.g., Rosaceae, Euphorbiaceae, Fagaceae, Oxalidaceae, and Violaceae) were also included. Numerous phylogenetic analyses were conducted with different combinations of outgroup and ingroup taxa. Based on these and published studies, primary outgroups used for full ingroup analyses were *Securidaca* and *Polygala* (Polygalaceae; GenBank accessions L01955 and L01945; sequences from Albert, Williams, and Chase, 1992).

Polymerase chain reaction (PCR) amplification of *rbc*L used primers described in Olmstead et al. (1992): a 5′ primer designed against the first 26 nucleotides of the *Nicotiana rbc*L gene, and a 3′ primer for a conserved region ≈100 bp 3′ of the *Nicotiana rbc*L stop codon. For a few recalcitrant samples, internal (sequencing) primers were used to amplify smaller pieces of the gene, which provided a complete overlapping sequence. A number of different amplification protocols were used over the course of these studies, involving various reaction conditions, polymerases, and thermal cyclers. Most were modifications of standard one-step PCR, but a two-step method was used for the most difficult samples. In this method (''booster PCR''; Ruano, Fenton, and Kidd, 1989), the first step was carried out in 20 μL total volume, used a variable amount of template DNA, 200 μmol/L dNTP, one unit *Taq* polymerase (TaKaRa or Pharmacia), and manufacturer-supplied 10× buffer, and consisted of a 2 min preheat at 95°C followed by 20 cycles of 94°C (1 min), 45°C (4 min), 60°C (1.5 min), with a final extension of 72°C (15 min). Primers, buffer, dNTP, and distilled water were then added to the tube to achieve a final volume of 50 μL, with final concentrations of 200 μmol/L dNTP and 0.4 μmol/L primers, and 1.25 units of *Taq* polymerase. Standard PCR conditions for the second cycle were a preheat of 94°C (1 min), and 25 cycles of 94°C (1 min), 56°C (1.5 min), 72°C (2 min), with a final extension of 72°C (15 min).

Amplified products were purified in one of several ways: (1) Wizard prep columns (Promega Corp., Madison, WI); (2) QIAGEN (Chatsworth, CA) columns; (3) GeneClean II (Bio 101, La Jolla, CA), following manufacturers' specifications. DNA sequencing was performed either (1) manually, using the snap-chill modification of the US Biochem Sequenase version 2.0 method, or (2) using an Applied Biosystems 373A automated sequencer. Sequencing primers for both strands were generally the ''universal'' *rbc*L primers provided by G. Zurawski (DNAX Corporation; described in Zurawski et al., 1981; Zurawski, Whitfield, and Bottomley, 1986). A number of sequencing primers were also designed from the published alfalfa *rbc*L sequence (Aldrich et al., 1986). Each was 20 nucleotides (nt) long unless indicated otherwise, and began at nucleotides 225 (21 nt long), 376 (17 nt), 552, 576 (19 nt), 787, 874, and 1076 of the coding strand, and at 325 (17 nt), 525 (13 nt), and 925 of the noncoding strand. These primers were designed for use in samples for which the Zurawski primers gave unsatisfactory results. All samples used in the full analysis were sequenced such that at least 80% of the sequence was confirmed either by reading the complementary strand or by obtaining overlapping reads on the same strand with a different primer.

As has been pointed out previously (e.g., Chase et al., 1993), alignment is unambiguous for the first 1428 nucleotides of the *rbc*L gene, but length variation occurs beyond this point. A great deal of length heterogeneity was observed in legume sequences. Consequently, for phylogenetic analyses only the first 1428 bp were used.

A number of different phylogenetic analyses were performed, all using equally weighted parsimony. The principal analysis included 84 taxa; subanalyses of papilionoid taxa were also conducted, as were analyses in which some taxa were added for which only incomplete sequences were available. Because all searches included too many taxa for efficient implementation of branch-and-bound or exhaustive searches, heuristic methods were used. These were implemented either in PAUP 3.1 (Swofford, 1993) or NONA (Goloboff, 1993). PAUP analyses used various combinations of taxon entry sequences and branch-swapping procedures in an effort to minimize problems of ''tree islands'' (Maddison, 1991). NONA analyses involved a two-stage procedure in which multiple (>1 000) runs were implemented with random taxon entry sequences with branch-swapping, but with a maximum of four trees held per run, followed by exhaustive branch swapping on the accumulated shortest trees from the first stage. Basic analyses included only bases 30–1428, excluding, at the 5′ end, primer sequences and, at the 3′ end, the length-variable region.

Robustness of clades was investigated primarily using the parsimony jackknife method of Farris et al. (in press), implemented using the program of Farris (1995). Other analyses were performed in PAUP using negative constraints or bootstrap options.

RESULTS

Parsimony analysis (using NONA) of the primary data set, consisting of 84 taxa and excluding the 3′ length-variable region of the gene, identified 184 most parsimonious trees of length (L) 1913, consistency index (ci) 0.35 (0.29 excluding autapomorphies), and retention index (ri) 0.52. The strict consensus of these trees was well-resolved (branches with solid bold lines in Fig. 1); resolution was least at the base of the Papilionoideae, where sequences from several members of the tribes Sophoreae

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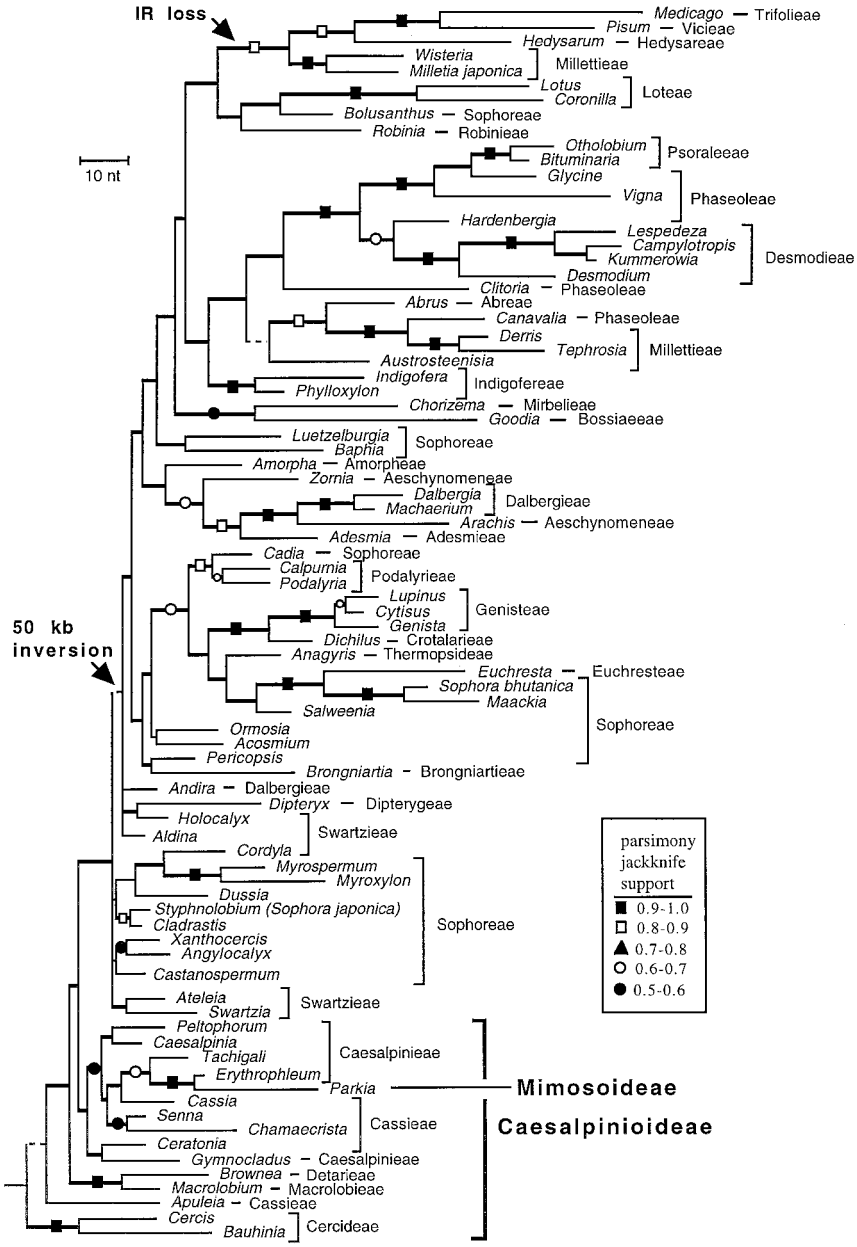


Fig. 1. One of 184 equally parsimonious trees for the full analysis of 84 taxa. Branch lengths are shown using ACCTRAN optimization (PAUP). Clades present in the strict consensus topology are indicated by bold lines, while those not present in the strict consensus tree are indicated (where long enough to be visible) by dashed lines. Parsimony jackknife results (50% support level or higher) are indicated on the tree, using the symbols shown. The distribution of two chloroplast genome structural characters, a 50-kb inversion, and the loss of the inverted repeat, are indicated.

and Swartzieae formed a polytomy. This was due primarily to the paucity of substitutions differentiating the *rbc*L genes of these taxa, as can be seen from the branch lengths of one of the most parsimonious trees identified in this analysis (Fig. 1). Short branches occurred elsewhere in this and the other most parsimonious trees, and were, not unexpectedly, associated with low levels of support as estimated using the parsimony jackknife procedure of Farris et al. (in press). Low support was obtained for many relationships, the exceptions being certain taxa belonging to the same or closely related tribes (e.g., *Cercis* and *Bauhinia*; *Macrolobium* and *Brownea*) and many relationships involving the more derived members of Papilionoideae. Sequences from ''higher'' papilionoid taxa appear to have accumulated considerably more substitutions than have those of other legumes, providing more characters for phylogeny reconstruction. Differences in branch lengths are presumably due to differences in rates of *rbc*L evolution among different taxa, as has been noted in other groups (e.g., Bousquet et al., 1992; Gaut et al., 1992).

The main analysis assumed monophyly of the Leguminosae, including as outgroups only two sequences from Polygalaceae that had been placed in previous more comprehensive analyses (Fernando et al., 1993; Chase et al., 1993; Morgan and Soltis, 1993; Morgan, Soltis, and Rob-

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ertson, 1994) as sister to the few legume sequences then available. Based on the strict consensus topology (Fig. 1), various combinations of several representative legume sequences, particularly several placed near the base of the tree, were included with a number of available nonlegume sequences in numerous parsimony analyses. These analyses consistently identified a monophyletic Leguminosae and placed Polygalaceae either alone or with Surianaceae as sister to legumes.

The polytomy at the base of the Papilionoideae in the legume strict consensus tree included taxa both with and without a 50-kilobase (kb) cpDNA inversion that marks a ''higher'' papilionoid group (Doyle et al., 1996). Some but not all most parsimonious trees were consistent with a monophyletic group of taxa possessing this inversion. Among the genera shown at the base of the subfamily in one such tree (Fig. 1), the presence/absence of the inversion has not been determined for *Sophora japonica* or *Dussia*, but the inversion is absent from all remaining taxa (Doyle et al., 1996). Of taxa predicted by this topology to possess the inversion, several have yet to be studied for the distribution of this character. Notable among these are the sole representatives of the tribes Adesmieae and Euchresteae, *Dipteryx* (which gave ambiguous results in Doyle et al., 1996), and two Sophoreae (*Sophora bhutanica* and *Salweenia*). A third member of Sophoreae, *Acosmium dasycarpum*, was reported in Doyle et al. (1996) as lacking the inversion, but belongs to the clade marked by the presence of this character. However, the specimen from which this sample was taken (and which was also used in the cpDNA gene/intron loss study of Doyle, Doyle, and Palmer, 1995) has more recently been identified as *Apuleia leiocarpa* (Caesalpinioideae: Cassieae). The *rbc*L sequence of this sample fell near the base of the entire tree, consistent with its lacking the inversion. The ''*Acosmium*'' sequence appearing in the *rbc*L tree was sampled from a newly collected and verified *Acosmium dasycarpum* sample; its condition for the 50-kb inversion is as yet unknown.

DISCUSSION

rbcL as a source of phylogenetic characters in Leguminosae—Including partial sequences and sequences from other investigators (e.g., Chase et al., 1993; Käss and Wink, 1995), *rbc*L data are now available from >100 of the ≈650 genera of the Leguminosae. The analyses discussed here include nearly all of the hypothesized major evolutionary lineages of the two largest and most diverse subfamilies, Caesalpinioideae and Papilionoideae. Trees produced as a result of these studies were generally well resolved, although many relationships were not supported by many characters. *rbc*L did not prove to be an ideal gene for studying relationships within lineages such as Caesalpinioideae or basal Papilionoideae, but did provide considerable data concerning relationships among more derived elements in the latter subfamily. Data were also insufficient for the construction of robust resolved hypotheses concerning the earliest radiations of the family. However, most major relationships (e.g., monophyly of Papilionoideae, nesting of Mimosoideae within Caesalpinieae) were largely insensitive to inclusion or exclusion of taxa, as can be seen, for example, by comparing Fig. 1 with a summary tree from an earlier smaller data set (Doyle, 1995, fig. 5). In general, then, despite limitations, data from *rbc*L sequences have proven useful as a means of constructing phylogenetic hypotheses for this large family. These hypotheses serve as a basis for discussion of legume taxonomy and evolution.

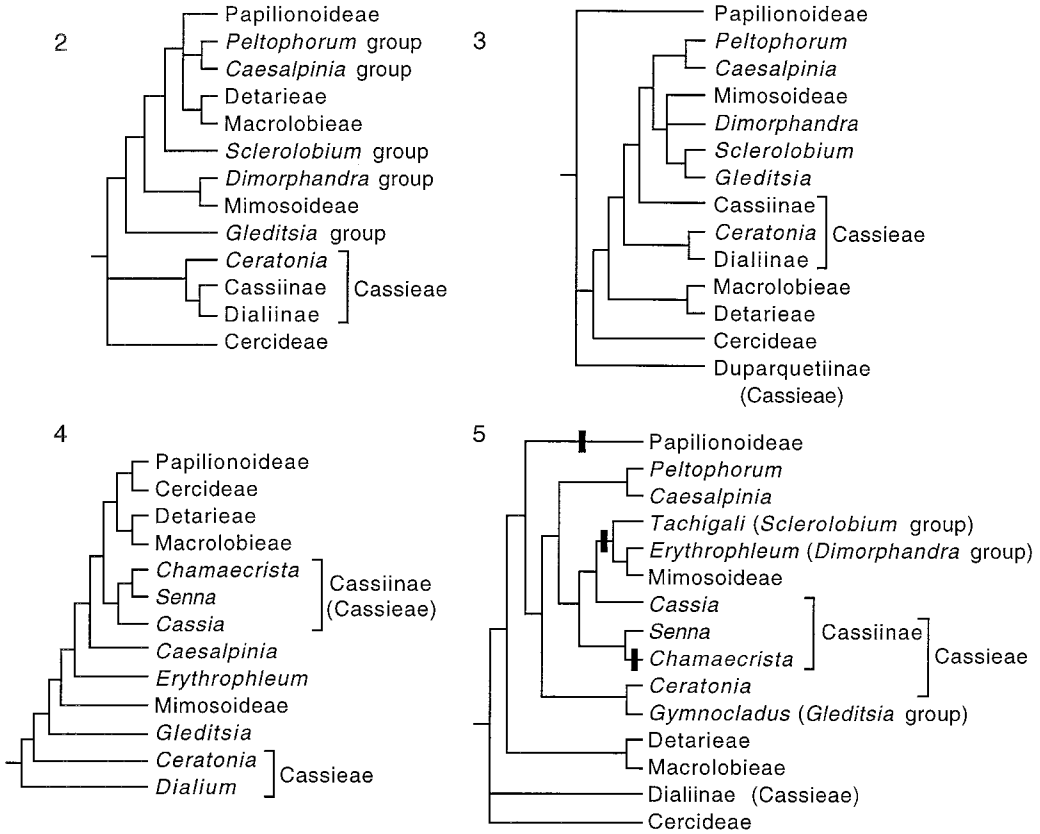
Familial relationships of Leguminosae—Although ''legumes'' have at different times been considered to comprise either a single family (Leguminosae or Fabaceae) with three subfamilies (Papilionoideae or Faboideae, Caesalpinioideae, and Mimosoideae; e.g., Polhill and Raven, 1981) or three separate families (Fabaceae, Caesalpiniaceae, and Mimosaceae; e.g., Cronquist, 1981), the naturalness of the group has rarely been questioned. As expected, monophyly of the family was uniformly supported in most parsimonious trees that included *rbc*L sequences from a number of families that have been suggested to be related to Leguminosae (e.g., Dickison, 1981; Chase et al., 1993; Morgan and Soltis, 1993; Morgan, Soltis, and Robertson, 1994).

Previous *rbc*L studies bearing on the issue of placement of Leguminosae within angiosperms have used only two complete published sequences (*Pisum*: Zurawski, Whitfield, and Bottomley, 1986; *Medicago*: Aldrich et al., 1986), and two unpublished partial sequences (*Albizia* and *Bauhinia*, both from J. Learn and M. T. Clegg, University of California, Riverside). The much greater sampling of legumes presented here is thus relevant in considering this question. Our results confirm the findings of previous studies (Chase et al., 1993; Morgan, Soltis, and Robertson, 1994) that the closest relationships of Leguminosae *rbc*L sequences are with Polygalaceae and Surianaceae particularly, and somewhat more distantly with *Quillaja* (an anomalous member of Rosaceae). Families previously suggested to have a close relationship to legumes, such as Sapindaceae (Dickison, 1981; Thorne, 1992), or Krameriaceae, Chrysobalanaceae, and Connaraceae (see Dickison, 1981), appear to be more distantly related.

In the Chase et al. (1993) study of *rbc*L sequences from 500 angiosperm taxa, a sequence obtained from sterile material identified as *Sargentodoxa* (Sargentodoxaceae) appeared as sister to the two papilionoid legume sequences. More recently, Hoot, Culham, and Crane (1995) have obtained an *rbc*L sequence from more confidently determined *Sargentodoxa*; in parsimony analyses this sequence does not fall anywhere near Leguminosae. The original ''*Sargentodoxa*'' sequence, when included with other legumes studied here, was placed as sister to *Millettia japonica*, a member of that portion of the tribe Millettieae that lacks the cpDNA inverted repeat (Lavin, Doyle, and Palmer, 1990; Liston, 1995) and is part of a clade with members of the ''temperate herbaceous group'' of tribes (Polhill, 1981e; see below).

Naturalness of legume subfamilies and relationships of caesalpinioid taxa—Whether recognized at the familial or subfamilial levels, it has been clear that while Papilionoideae and Mimosoideae likely are each monophyletic groups, Caesalpinioideae is probably not natural. Rather, it has been considered para- or polyphyletic, including genera comprising sister groups to the other two

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Figs. 2–5. Hypotheses of relationships among Papilionoideae, Mimosoideae, and principal groups of Caesalpinieae. **2.** Cladistic representation of the hypothesis of Polhill, Raven, and Stirton (1981), based on morphological considerations. **3.** Strict consensus topology of Chappill's (1995) preliminary cladistic analysis, largely using morphological data. **4.** Strict consensus topology from the morphological cladistic analysis of Tucker and Douglas (1994), which included numerous floral ontogenetic characters. **5.** Strict consensus topology from *rbc*L study; black bars represent parsimonious optimization of the character ''nodulation.'' Trees in Figs. 3 and 4 are reduced from the original analyses to show only taxa discussed here. In all trees, genera, generic groups, or subtribes not otherwise identified to tribe are members of Caesalpinioideae, tribe Caesalpinieae.

subfamilies and also elements basal in the entire family (Fig. 2; Polhill, Raven, and Stirton, 1981). These hypotheses are to a considerable extent supported by the *rbc*L data. Sequences of the well-sampled Papilionoideae formed a monophyletic group, though weakly supported; Mimosoideae were represented in this study by only a single complete sequence (*Parkia*), but the taxa sampled here, representing in at least some analyses four of the five tribes of the subfamily, uniformly appeared as sister taxa in all trees. In contrast, Caesalpinioideae sequences occurred in four distinct places in topologies of most parsimonious trees: (1) *Bauhinia* and *Cercis* (Cercideae), and (2) *Apuleia* (Cassieae: Dialiinae) formed clades that were part of the basal trichotomy in strict consensus trees; (3) *Macrolobium* (Macrolobieae) and *Brownea* (Detarieae) formed a clade that was the next dichotomy in the family; (4) all remaining Caesalpinioideae formed a clade that also included Mimosoideae (Fig. 1).

Cercideae is a small tribe of only five genera, and is considered a natural group (Wunderlin, Larsen, and Larsen, 1981) that constitutes one of the three major radiations of Leguminosae, and is marked by a number of morphological differences from the remainder of the family (Polhill, Raven, and Stirton, 1981). Strongly supported monophyly of sequences representing its two subtribes is therefore not surprising. Similarly, Detarieae and Macrolobieae (formerly Amherstieae; Breteler, 1995) are considered very closely related. Fragmentary unpublished *rbc*L sequences (nucleotides 1–398) exist in GenBank (accessions L08477–80; H. N. Poinar, G. O. Poinar, Jr., and R. J. Cano) for four extant and putative fossil species of *Hymenaea*, a member of the *Amherstia* group of Detarieae (Breteler, 1995). Inclusion of any of these sequences in the full data matrix consistently placed it as part of this Detarieae-Macrolobieae clade, further supporting the naturalness of this group of taxa.

In contrast, Cassieae was polyphyletic in the *rbc*L analysis, contradicting the notion that this tribe represents a natural group derived from a second major radiation in the family (Polhill, Raven, and Stirton, 1981; Fig. 2). *Apuleia*, the single representative included in the full analysis from the largest subtribe, Dialiinae, was placed at or near the base of the family. A partial sequence from *Dialium* was sister to the *Apuleia* sequence, suggesting that this subtribe forms an evolutionary lineage separate from the remainder of the family. The remaining Cassieae were intermixed with members of the caesalpinioid tribe Caesalpinieae in the large clade that also included Mimosoideae. *Ceratonia* (subtribe Ceratoniinae), considered by Polhill, Raven, and Stirton (1981) to be the basal member of a natural Cassieae, was instead relatively basal in this latter clade, joining *Gymnocladus* (*Gleditsia*

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group of Caesalpinieae). Three genera of subtribe Cassiinae (*Cassia, Senna,* and *Chamaecrista*) appeared elsewhere in the Caesalpinieae/Mimosoideae clade, but only the latter two formed a clade. Absence of close relationship among subtribes of Cassieae is perhaps not surprising; Irwin and Barneby (1981), though recognizing a single tribe, did so with considerable hesitation, and apparently only because of pressure from their colleagues not to increase the number of tribes in the subfamily. Their view was that the subtribes could as easily be elevated to tribal status, though they did not go so far as to suggest that these taxa are unrelated. The failure of the three Cassiinae genera to form a clade is more surprising; support was not particularly strong even for the *Senna–Chamaecrista* grouping. These three genera were until relatively recently considered a single genus, *Cassia,* and they share a number of distinctive morphological characters (Irwin and Barneby, 1981).

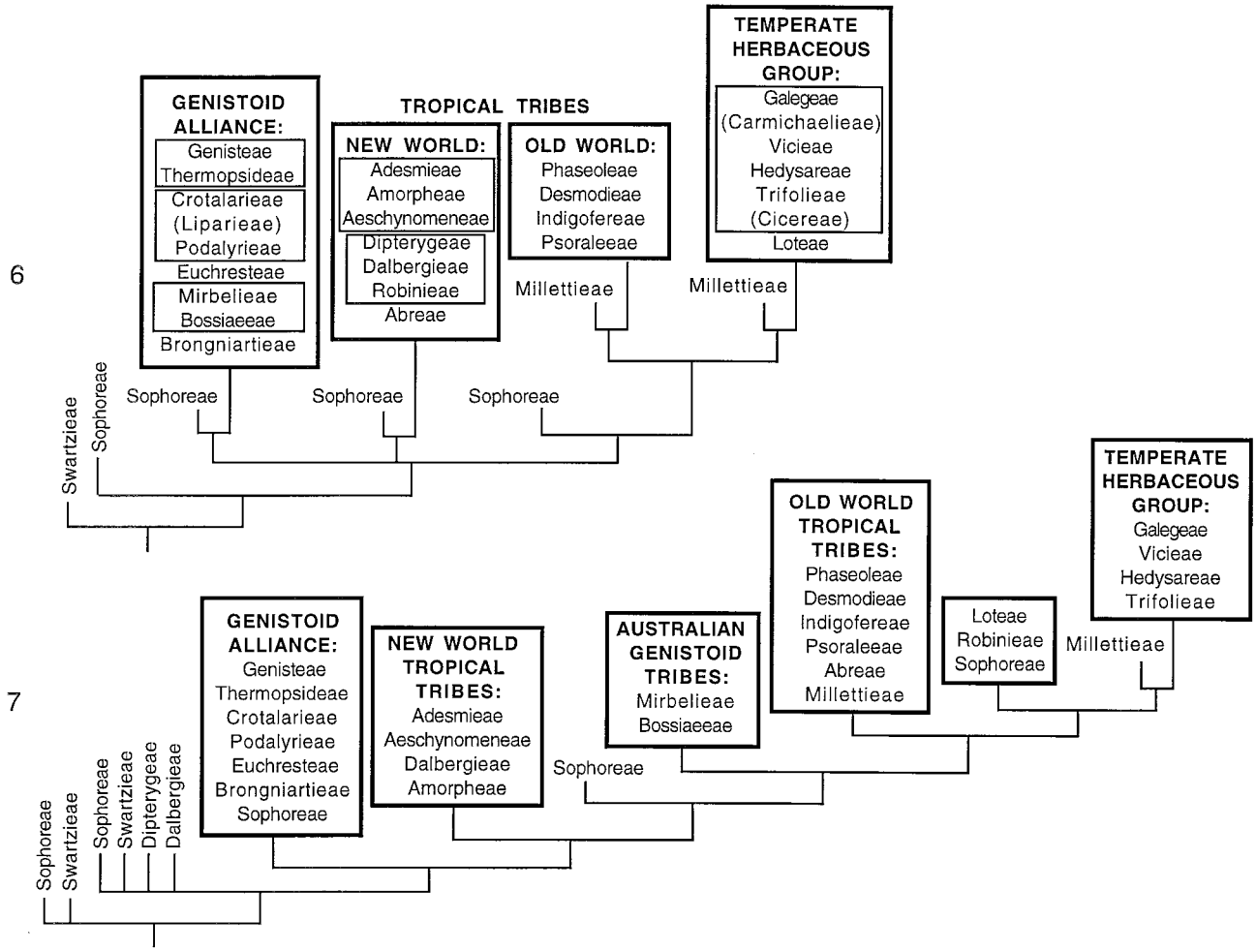
Caesalpinieae has been thought to be paraphyletic, including basal members of the third, and largest, radiation in the family, which produced a monophyletic group that comprises Papilionoideae and Mimosoideae as well as Detarieae/Macrolobieae (Polhill, Raven, and Stirton, 1981; Fig. 2). Although monophyly of this larger group of taxa is not supported by the *rbc*L data (Fig. 1), Caesalpinieae is indeed paraphyletic, with members scattered throughout the clade that also includes Cassiinae, *Ceratonia,* and Mimosoideae (CCCM clade). Polhill, Raven, and Stirton (1981) suggested that the *Dimorphandra* generic group of Caesalpinieae was sister to Mimosoideae; in the *rbc*L tree, *Erythrophleum* (*Dimorphandra* group) was strongly supported as sister to *Parkia,* the sole mimosoid from which a complete sequence was available. The partial sequences from other mimosoid tribes (Mimoseae: *Leucaena*; Ingeae: *Albizia, Calliandra*; Acacieae: *Acacia*), when included in analyses, consistently joined *Parkia,* suggesting both that Mimosoideae is monophyletic and that this is the likely placement of the subfamily. The member of the *Sclerolobium* group of Caesalpinieae sampled here (*Tachigali*) was sister to the *Erythrophleum*/Mimosoideae clade instead of being sister to Papilionoideae, Detarieae/Macrolobieae, and the *Peltophorum* and *Caesalpinia* groups of Caesalpinieae, as was suggested by Polhill, Raven, and Stirton (1981; Fig. 2). Together, *Tachigali, Erythrophleum,* and Mimosoideae have in common participation in symbiotic atmospheric nitrogen fixation (nodulation), a trait that distinguishes them from the remaining members of the CCCM clade (see below for further discussion).

The *Gleditsia* group of Caesalpinieae, to which *Gymnocladus* belongs, was considered a basal member of the main radiation of Leguminosae by Polhill, Raven, and Stirton (1981; Fig. 2), primarily on account of the reduced unisexual flowers found in its two genera. In the *rbc*L tree, *Gymnocladus* was sister to *Ceratonia,* another genus with unisexual flowers that has been considered primitive, and this clade was basal within the CCCM clade. Thus the two taxa with unisexual flowers are nested between Detarieae/Macrolobieae and the remainder of the CCCM clade, groups that typically have well-developed, perfect flowers, suggesting that floral reduction is potentially a synapomorphy for these two taxa rather than being a plesiomorphic condition. These results are also consistent with the suggestion by Herendeen, Crepet, and Dilcher (1992) that ideas about ''primitive'' legumes need to be reevaluated. Neither *Ceratonia* nor *Gymnocladus* appear early in the fossil record, nor do they appear to be early divergences based on *rbc*L results.

Paraphyly of basal papilionoid tribes—Swartzieae and Sophoreae have been suggested to represent ''primitive'' elements within Papilionoideae. Indeed, members of Swartzieae were until fairly recently classified as Caesalpinioideae (Cowan, 1981), and this tribe contains at least some taxa that may eventually be returned there (Polhill, 1994). Sophoreae has been recognized as ''a tribe of convenience between Caesalpinioideae and the bulk of Papilionoideae, sharply defined from neither'' (Polhill, 1981f, p. 213). Many of its genera have been thought to share more recent common ancestry with other papilionoid tribes than with other genera classified as Sophoreae. Polhill (1981e, f) depicts these relationships as a series of concentric circles representing grades of morphological specialization; he derives his much more frequently cited tree diagram from this ''target'' diagram with an explicit warning against overinterpretation. Swartzieae are at the center of the ''target'' diagram, with the various generic groups of Sophoreae occupying positions in the next several rings, and the ''advanced'' papilionoid tribes are at the periphery. The clear implication (Fig. 6) is that both Sophoreae and generic groups within that tribe are para- or polyphyletic, with individual genera being part of various radiations leading to the more specialized members of the subfamily. Achieving an understanding of the phylogenetic relationships of genera classified in these tribes is thus likely to be an important step in elucidating the evolutionary history of the subfamily. A recent morphological cladistic analysis of Swartzieae and Sophoreae (Herendeen, 1995) suggested that these two tribes are completely interdigitated, comprising several distinguishable lineages that include members of both tribes. Similarly, Polhill's 1994 taxonomic summary of the family transferred some genera previously classified as Sophoreae (e.g., *Ateleia*) to Swartzieae. Chappill's (1995) preliminary cladistic analysis of the family also showed Sophoreae to be paraphyletic, placing the only two Swartzieae sampled in a clade at the base of the subfamily. The 50-kb cpDNA inversion, which distinguishes most Papilionoideae from all Mimosoideae and Caesalpinioideae, is found in some but not all genera of Swartzieae and Sophoreae (Doyle et al., 1996), further suggestive of the unnaturalness of these tribes.

The *rbc*L topology is consistent with para- or polyphyly of Swartzieae and Sophoreae (Figs. 1, 7). Eight of the 18 Sophoreae and all five of the Swartzieae sampled here were part of the unresolved basal group of the Papilionoideae in the strict consensus *rbc*L tree (Fig. 1), in agreement with hypotheses that these tribes include basal elements of the subfamily. Even considering only these taxa and excluding more distantly placed Sophoreae and Swartzieae, neither tribe appears to be monophyletic. Results support the recent transfer of *Ateleia* from Sophoreae to Swartzieae (Polhill, 1994), while in at least one case a genus of Swartzieae (*Cordyla*) is suggested to be more closely related to Sophoreae (*Myroxylon* and *Myrospermum*) than to other Swartzieae.

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Figs. 6–7. Tribal relationships in Papilionoideae. **6.** Radiation of papilionoid tribes, modified from Polhill (1981e). Tribes not included in *rbc*L analyses are shown in parentheses. Note paraphyly (or polyphyly) of Sophoreae and Millettieae. Within the Genistoid Alliance, three geographic groups are indicated by boxes (see text for discussion). Boxed tribes in the Temperate Herbaceous Group are those lacking the chloroplast DNA inverted repeat, a trait shared with some Millettieae (Lavin, Doyle, and Palmer, 1990; Liston, 1995). **7.** Tribal relationships inferred from *rbc*L strict consensus tree. Group names are retained from Fig. 7 to indicate predominant membership (e.g., ''New World Tropical Tribes'' is used for the group including Phaseoleae, despite the fact that both New World and Old World Millettieae genera belong to this clade).

Several additional Sophoreae genera are scattered among other Papilionoideae in a pattern that agrees with the conclusions based on the distribution of the 50-kb inversion. Indeed, some but not all most parsimonious *rbc*L trees were consistent with the inversion being a unique event marking a clade that includes several genera of Sophoreae and Swartzieae as well as all those elements of Papilionoideae that have been considered ''advanced'' (Polhill, 1981e). Relationships of some Sophoreae with members of these derived papilionoids in some cases confirm previous hypotheses. *Cadia* has been suspected to be related to Podalyrieae (van Wyk and Schutte, 1995), while *Calpurnia*, now placed in Podalyrieae (Polhill, 1994), was until recently placed in Sophoreae. These two genera and *Podalyria* formed a well-supported clade here. The separation of *Sophora japonica* from other members of *Sophora* parallels results of the independent *rbc*L analysis of Käss and Wink (1995; whose sampling of the genus was somewhat different than ours), and also agrees with its segregation (as *Styphnolobium*; Sousa and Rudd, 1993) from *Sophora* s.s. The close relationship between this taxon and *Cladrastis* suggested by the *rbc*L tree has not previously been hypothesized, but the separation of the latter genus from *Maackia*, to which it has been considered closely related, is consistent with the distribution of the 50-kb inversion (Doyle et al., 1996). *Maackia* joins a species of *Sophora* s.s. in a clade with Genisteae and allies (see below); this placement for *Sophora* s.s. is in agreement with *rbc*L results of Käss and Wink (1995). Our *rbc*L results also suggest relationships of some genera, such as *Salweenia* (*Sophora* group), whose affinities have simply been unknown (Polhill, 1981f). These results also suggest that current informal generic groupings within Sophoreae appear in most cases to be artificial. Of the six groups recognized by Polhill (1994), four were represented by more than one genus here, and none of these formed monophyletic groups in the *rbc*L consensus tree, although some pairs of genera from the same group did form clades (e.g., *Myrospermum* and *Myroxylon*).

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"Higher" Papilionoideae—The first of the groups stemming from the radiations envisioned by Polhill (1981e, 1994; Polhill, Raven, and Stirton, 1981; Figs. 6, 7f) is a ''Genistoid Alliance'' that includes (1) the predominantly northern hemisphere tribes Genisteae and Thermopsideae; (2) the largely southern African tribes Podalyrieae, Crotalarieae, and Liparieae; and (3) the Australian Mirbelieae and Bossiaeeae. A second major radiation comprises a group of tropical tribes that includes at its base a ''diffuse woody part'' consisting of such tribes as Dalbergieae, Dipterygeae, Millettieae (called ''Tephrosieae'' in Polhill, 1981e), and Robinieae. The more derived elements of this second major group are split geographically into predominantly Old World (Phaseoleae, Desmodieae, Indigofereae, and Psoraleeae) and New World (Adesmieae, Amorpheae, and Aeschynomeneae) tribes thought to be derived from ancestors that would be classified as Millettieae. A further radiation from millettioid progenitors produced a number of generally temperate and herbaceous tribes allied to Galegeae. Several major clades were identified by the *rbc*L analysis within the group marked by the 50-kb inversion, some of which corresponded to groups of tribes derived from these hypothesized radiations (Fig. 6).

"Genistoid Alliance"—The bulk of this group was represented by a clade in the *rbc*L strict consensus tree comprising sequences from representatives of Podalyrieae, Crotalarieae, Genisteae, and Thermopsideae (Figs. 1, 7). The two Australian ''genistoid'' tribes, Mirbelieae and Bossiaeeae, however, did not join this clade. Furthermore, within the clade there was no dichotomy between African and Northern Hemisphere taxa. A strongly supported monophyletic Genisteae included *Lupinus,* as was also found by Käss and Wink (1995). Also as in their parsimony tree, Genisteae, a Northern Hemisphere tribe, was sister to *Dichilus,* a member of the largely southern African tribe Crotalarieae. The representative of the remaining Northern Hemisphere tribe, Thermopsideae (*Anagyris*), fell outside this strongly supported clade, and was instead sister to a group of Asian genera that included three Sophoreae of the *Sophora* group (*Maackia,* a Himalayan *Sophora* species, and the Tibetan endemic *Salweenia*), and also *Euchresta* (of the monogeneric Euchresteae), which has been considered to be related to the *Sophora* group (Polhill, 1994). In the Käss and Wink (1995) tree, Thermopsideae is paraphyletic, with a clade of three *Sophora* species (*S. davidii, S. jaubertii,* and *S. flavescens*) sister to the clade formed by *Anagyris* and *Piptanthus,* and with this entire group sister to *Thermopsis* and *Baptisia.* The placement of the ''genistoid'' Sophoreae sampled here is consistent with their topology, with this group sister to *Anagyris* (the only included member of Thermopsideae). The remaining sampled Southern Hemisphere genistoid tribe, Podalyrieae, joined by two sophoroid genera (*Cadia* and *Calpurnia,* discussed above) was sister to the clade that included Thermopsideae, Genisteae, and Crotalarieae. This is in contrast to the Käss and Wink (1995) tree, in which *Podalyria* was sister to Genisteae (*Cadia* and *Calpurnia* were not included in their study). In the strict consensus tree, this whole group was joined by several other Sophoreae and also by *Brongniartia* (Brongniartieae). The affinities of the latter tribe have been thought to be genistoid, though perhaps with the Australian tribes Bossiaeeae and Mirbelieae (Crisp and Weston, 1987), which as noted did not join the genistoid group here.

Advanced tropical tribes and their woody relatives—In Polhill's (1981e) commonly cited tree diagram, Dalbergieae is placed as an early offshoot of Sophoreae, and is not part of the main radiation of the remaining Papilionoideae. The tribe is likely paraphyletic (Polhill, 1981c; Lima, 1991), its members sharing the diagnostic but evolutionarily labile character of indehiscent fruits. Of the three members of Dalbergieae included in the *rbc*L analysis, *Andira,* the sole member sampled from the *Andira* generic group (Polhill, 1994), was part of the unresolved polytomy at the base of the Papilionoideae in the strict consensus tree. In contrast, the remaining two genera sampled (*Machaerium* and *Dalbergia,* both members of the *Dalbergia* group), were strongly supported as sister taxa and formed part of a clade that included sequences from members of three other tribes, Amorpheae, Aeschynomeneae, and Adesmieae. According to Polhill (1994), chemosystematic considerations suggest a link between Aeschynomeneae and Dalbergieae, and placement of these particular Dalbergieae genera with Aeschynomeneae is also supported by several morphological characters, such as helicoid cymose inflorescences, wood with storied rays, forward leaflet nyctinasty, and paripinnate leaves (Lavin and Sousa, 1996; M. Lavin, Montana State University, personal communication). Amorpheae, though not sharing these characters, appears to be related to this clade in such characters as ancestral base chromosome number (M. Lavin, personal communication). Aeschynomeneae, Adesmieae, and Amorpheae comprise the New World advanced tropical tribes of millettioid descent envisioned by Polhill (Fig. 6), and their separation from Old World tropical elements is supported by palynological characters. Polhill elsewhere (1981b) suggests a relationship of Adesmieae with Robinieae, Dalbergieae, Amorpheae, and particularly with Aeschynomenae. Paraphyly of the Aeschynomeneae, as supported here, is not unprecedented; *Zornia* and *Arachis* represent two different subtribes (Poiretiinae and Stylosanthinae, respectively) that did not form a monophyletic group in Chappill's (1995) preliminary cladistic analysis of the family. Although Polhill (1981e) suggested that these tribes are derived from millettioid ancestors, there is no indication of a relationship with any Millettieae in the current *rbc*L data set or with other sequences from that tribe (T. Kajita, H. Ohashi, J. L. Doyle, and J. J. Doyle, unpublished data).

Even so, Millettieae does not appear to be monophyletic, based on the *rbc*L results. This comes as no surprise given its position relative to other taxa in treatments of Polhill (1981e; Fig. 6) and Geesink (1981, 1984), its polymorphism for the presence of the IR (Lavin, Doyle, and Palmer, 1990; Liston, 1995), and the placement of its genera in phylogenetic analyses of such putatively derived tribes as Robinieae (Lavin, 1995) and Phaseoleae (Doyle and Doyle, 1993). Apart from the two Millettieae that lack the IR (see below), three other genera of this tribe were included. *Tephrosia* and *Derris* were strongly supported as sister taxa, while *Austrosteenisia* was not

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sister to these taxa but was placed more basally in the same large clade. *Austrosteenisia* species have until relatively recently (Geesink, 1984) been classified as *Kunstleria*, a divergent and relatively anomalous member of the tribe (Ridder-Numan and Kornet, 1994), so its placement separate from other Millettieae is perhaps not surprising. Ongoing *rbc*L studies of this large tribe (43 genera in Polhill, 1994) suggest that additional genera can be placed either with those lacking the IR or in the *Derris* clade, but that at least some taxa have affinities elsewhere (T. Kajita, H. Ohashi, J. L. Doyle, and J. J. Doyle, unpublished data).

The clade that contains these Millettieae also includes Phaseoleae, the largest legume tribe (83 genera in Polhill, 1994). This tribe was presumed to be para- or polyphyletic by Lackey (1981), a hypothesis supported by cpDNA restriction site mapping studies of the IR (Doyle and Doyle, 1993). That conclusion is further bolstered here, with various Phaseoleae appearing throughout this large clade. Perhaps the most unexpected result of the cpDNA RFLP (restriction fragment length polymorphism) study was the suggestion that Desmodieae, a morphologically very distinctive tribe, was nested within what had been considered core elements of the Phaseoleae (e.g., subtribes Glycininae, Phaseoleae, and Kennediinae). These groups were all included in the *rbc*L analysis, and Desmodieae sequences formed a monophyletic group that was again nested within core Phaseoleae, sister to subtribe Kennediinae (*Hardenbergia*). Ongoing *rbc*L studies in Desmodieae, combined with its polymorphism for the intron in the chloroplast gene *rpl2* (Bailey et al., 1995 and unpublished data), suggest that the tribe is not monophyletic as currently circumscribed, with subtribe Bryinae (formerly placed in Aeschynomeneae; Kajita et al., 1996) and some genera of subtribe Lespedezinae likely having affinities elsewhere (Kajita et al., 1996 and unpublished data). Psoraleeae, another of the tribes considered by Polhill (1981e) to be derived from millettioid ancestors, was not included in the earlier restriction site study. Two genera of that tribe formed a well-supported clade in the *rbc*L tree, and this appeared as sister to *Glycine* (Phaseoleae subtribe Glycininae) in a clade that also included *Vigna* (Phaseoleae subtribe Phaseolinae). This would have been surprising, had it not paralleled the results of a study of the nuclear copy of the mitochondrial gene encoding cytochrome oxidase subunit II (*coxII*) in Phaseoleae and allies (K. Song, J. D. Palmer, and J. J. Doyle, unpublished data). In that study, as here, Psoraleeae sequences joined Glycininae in a well-supported clade nested within Phaseoleae/Desmodieae. Morphologically, Psoraleeae are typically trifoliolate, like most Phaseoleae, and the tribe shares its pseudoracemose inflorescence with Phaseoleae, Desmodieae, Abreae, and most Millettieae (Tucker, 1987a, b). The separation of Psoraleeae from Amorpheae is not surprising, because a major character once considered to unite them, the discontinuity plate (a feature of floral vasculature), has more recently been shown to be nonhomologous in the two tribes (Grimes, 1986).

Taxa possessing pseudoracemes formed a monophyletic group in some and perhaps all most parsimonious *rbc*L trees. *Austrosteenisia* (Millettieae) is placed either basal to this clade or as sister to one of the two subclades within it (as in Fig. 1). The infloresence of this genus is described as a ''terminal panicle'' by Geesink (1984), but may in fact be a pseudoraceme (M. Lavin, personal communication). Relationships of one of the pseudoraceme-bearing tribes, the monogeneric Abreae, have been considered questionable, though its affinities could be with African Millettieae (Polhill, 1981a), a placement consistent with our findings. In contrast, the *rbc*L study of Käss and Wink (1995) placed *Abrus* with taxa of the temperate herbaceous group, an unusual result given the presence of the cpDNA IR in *Abrus* (Lavin, Doyle, and Palmer, 1990). Oddly, their trees grouped *Glycyrrhiza* (Galegeae), a genus known to lack the IR (Lavin, Doyle, and Palmer, 1990), with Phaseoleae. Inclusion in our analysis of the sequences of these two genera, as reconstructed from data given in their paper (and more recently from their GenBank submissions), separated the two *Abrus* sequences, and placed their *Glycyrrhiza* sequence sister to our *Abrus* sequence. Because we were concerned that our sequence might possibly represent a contaminant or other error (see ''PCR nightmares'' of Olmstead and Palmer, 1994), we generated a second sequence from a new DNA isolation of our *Abrus* plant, and also produced a sequence from *Glycyrrhiza*. Phylogenetic analyses placed our two *Abrus* sequences together, and placed our *Glycyrrhiza* sequence with their *Abrus* sequence in the clade lacking the IR. These results seem best accounted for by a simple switch of *Glycyrrhiza* and *Abrus* sequences by Käss and Wink (1995).

Placement of Indigofereae as sister to the large millettioid/phaseoloid clade was not strongly supported, but is consistent with the suggestion of Polhill (1981e) that the tribe is derived from millettioid ancestors. Schrire (1995) also suggests a possible affinity with Millettieae, and uses a member of Millettieae (*Philenoptera*) as an outgroup in his cladistic analysis of Indigofereae, the other outgroup being the more distantly related *Baphia* (Sophoreae). The strongly supported relationship between the two Indigofereae genera sampled here is of interest because the Madagascan endemic *Phylloxylon*, though traditionally placed in this tribe, has such a highly modified morphology (e.g., it is leafless, with cladodes) that its placement has at times been questioned, and is at best thought to be sister to the remaining Indigofereae (Schrire, 1995).

The temperate herbaceous (TH) group—The final radiation of Papilionoideae envisioned by Polhill (1981e, 1994; Fig. 6) comprises Galegeae, Carmichaelieae, Cicereae, Hedysareae, Vicieae, and Trifolieae, as well as Loteae (including the formerly recognized Coronilleae; Polhill, 1981d, 1994). All but the last lack one copy of the cpDNA IR, a characteristic not shared by other legume tribes, with the exception of a few members of Millettieae (Lavin, Doyle, and Palmer, 1990; Liston, 1995). This character was included in Chappill's (1995) preliminary phylogenetic analysis of the family, but was insufficient either to overcome the many morphological similarities shared by the IR-lacking members of the TH group and Loteae or to unite the IR-lacking TH group and those Millettieae that also lack the IR. In contrast, the strict consensus *rbc*L tree does contain a well-supported IR-lacking clade (Figs. 1, 7), in which two Millettieae (*Wisteria* and *Millettia japonica*; the latter con-

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sidered by Geesink [1984] as a species of *Callerya*) form a clade sister to Vicieae, Trifolieae, and Hedysareae. Other TH IR-lacking tribes were not included in the full analysis, but two Galegeae sequences also joined this clade (a *Caragana* sequence reconstructed from Käss and Wink [1995], and the *Glycyrrhiza* sequence produced to confirm our *Abrus* results, as described above). Two Loteae formed a monophyletic pair that was part of a clade sister to this IR-lacking group. Sister to Loteae was *Bolusanthus* (Sophoreae), with *Robinia* (Robinieae) diverging as a basal dichotomy, but these latter relationships, as well as the grouping of this clade with the IR-lacking group, were not strongly supported. These results suggest, however, that TH-group features shared by Loteae, such as the loss of a pulvinus, herbaceous habit, and open vascular system, are indeed parallelisms, but could perhaps represent a homoplastic tendency (e.g., Sanderson, 1991) in two groups sharing some common ancestry. Of course, the *rbc*L results do not serve as a truly independent test of the hypothesis that the IR loss occurred only once in legume phylogeny, because both *rbc*L and the IR are part of what is a single effective historical unit, the chloroplast genome.

Sister to the combined TH and Phaseoleae clades are genera representing the two Australian genistoid tribes (Mirbelieae and Bossiaeeae). The relationships of these Australian tribes have been considered to be uncertain (Crisp and Weston, 1995; van Wyk and Schutte, 1995), and remain so, their placement in this clade being only weakly supported.

Evolution and diversification of Leguminosae: implications for the evolution of nodulation—The *rbc*L results include a basal trichotomy among three groups of very unequal size. The first of these is the small tribe Cercideae, with only five genera and fewer than 275 species, of which ≈250 are in *Bauhinia.* The second element of the trichotomy is subtribe Dialiinae of the caesalpinioid tribe Cassieae, a group that includes only 13 genera and ≈60 species. The third clade includes the remainder of the family, over 600 genera. The *rbc*L strict consensus tree (Figs. 1, 5) is in this regard thus quite similar to the hypothesis of Polhill, Raven, and Stirton (1981; Fig. 2), which also suggests an evolutionary trichotomy among Cercideae, elements of Cassieae (albeit including subtribes placed elsewhere based on *rbc*L), and the remainder of the family. Cercideae is a distinctive tribe (e.g., Polhill, 1994), with unique leaf and seed features and a putatively primitive base chromosome number ($x = 7$), which could represent plesiomorphic features (or, alternatively, autapomorphies). Genera of Dialiinae are unusual in lacking the typical legume fruit, having instead drupes or samaras (Irwin and Barneby, 1981). The lack of typical legumes could be a plesiomorphy, though drupaceous or samaroid fruits occur as apomorphic modifications elsewhere in the family, for example, in Dalbergieae (Papilionoideae) and Detarieae (Caesalpinioideae). A less ambiguous potential symplesiomorphy is the absence, from the wood of both Cercideae and Dialiinae, of the vestured pits characteristic of Leguminosae (Wheeler and Baas, 1992). It is therefore perhaps not unreasonable that Cercideae and/or Dialiinae could represent early divergences in the family, but the fossil record of Leguminosae offers no support for this hypothesis. While many modern groups are known from the Eocene, neither Cercideae nor Dialiinae has been identified from that period (Herendeen, Crepet, and Dilcher, 1992). Two other Cassieae subtribes, Duparquetiinae and Labicheinae, not sampled here, also possess wood lacking vestured pits, and one of these, Duparquetiinae, is part of a trichotomy at the base of the family in the cladistic analysis of Chappill (1995; Fig. 3). It is possible that these Cassieae subtribes, or one or more of the four unsampled and poorly known monogeneric groups of Caesalpinieae (*Acrocarpus, Poeppigia, Pterogyne,* and *Orphanodendron* groups; Polhill, 1994) could represent other early divergences in the family.

In the largest portion of the basal trichotomy, Macrolobieae/Detarieae are sister to the remaining Leguminosae. These tribes have in common with Cercideae and Dialiinae their inability to participate in symbiotic nitrogen fixation with soil bacteria, the syndrome known as nodulation. Of the families that are suggested as the closest relatives of Leguminosae, based either on *rbc*L data or in previous morphologically based hypotheses, none are known to nodulate (Soltis et al., 1995). This suggests that absence of nodulation is the plesiomorphic condition in Leguminosae, and that the origin of nodulation in the family occurred after the divergence of Dialiinae, Cercideae, and Macrolobieae/Detarieae from the remainder of the family. Nodulation occurs almost universally in Papilionoideae and Mimosoideae (though not in *Parkia*), but is sporadic in Caesalpinioideae (Faria et al., 1989). In the CCCM clade, the basalmost members are all taxa that are not known to nodulate (e.g., *Gymnocladus, Ceratonia,* and *Peltophorum*). *Erythrophleum* (*Dimorphandra* group of Caesalpinieae) is known to nodulate, one reason for its having been suggested as a link to Mimosoideae. *Tachigali,* which is sister to *Erythrophleum*-Mimosoideae, also nodulates, as do the other members of the *Sclerolobium* group of Caesalpinieae, to which it belongs. Of the three genera segregated from *Cassia* s. l. by Irwin and Barneby (1981), only *Chamaecrista* nodulates. Given the *rbc*L topology, particularly the sister group relationship of *Chamaecrista* and the nonnodulating *Senna,* the most parsimonious mapping of the character ''nodulation'' for these taxa is as a parallel gain, once in the common ancestor of Mimosoideae, *Erythrophleum,* and *Tachigali,* and a second time in *Chamaecrista* (Fig. 5). However, the weak support for several of these relationships, coupled with the paraphyly of Cassiinae and the closer than expected affinities between its genera and Caesalpinieae/Mimosoideae, suggests that a single origin of nodulation in this group might not be unreasonable.

Given the topology of the *rbc*L tree, nodulation in Papilionoideae is most parsimoniously hypothesized as occurring in parallel with the origin or origins of the trait in Caesalpinioideae/Mimosoideae (Fig. 5). Well over 90% of Papilionoideae thus far studied are thought to nodulate (Faria et al., 1989). However, among the basal or near basal Papilionoideae included here, nodulation is not universal. While *Swartzia, Ateleia, Xanthocercis,* and *Andira* are known to nodulate, *Castanospermum, Cladrastis, Myroxylon, Cordyla, Dipteryx,* and *Holocalyx* apparently do not, while for other taxa evidence is either

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lacking (*Angylocalyx, Dussia, Myrospermum, Aldina*) or is ambiguous (*Styphnolobium*) (Faria et al., 1989). Apart from a few genera whose nodulation status is unknown (*Salweenia, Brongniartia, Euchresta, Phylloxylon, Austrosteenisia,* and *Campylotropis*) or for which data are conflicting (*Maackia, Acosmium*), all of the remaining Papilionoideae sampled here are known to nodulate, with the exception of *Luetzelburgia* (Faria et al., 1989). The sporadic distribution of nodulation at the base of the subfamily, the paucity of records for some basal taxa, and the lack of robust resolution in that part of the *rbc*L tree combine to make it impossible to identify the point of a single origin of nodulation in Papilionoideae. Potentially, the ancestor of all Papilionoideae was capable of nodulating, but some taxa have lost the ability to do so; certainly this seems plausible for *Luetzelburgia,* which does not nodulate but is embedded within a group of genera nearly all of which are known to do so.

Nodulation is an energetically expensive trait, and often does not occur even in legume species capable of nodulation, either in stressed plants or in plants given adequate accessible nitrogen (Sprent, 1994). Moreover, single-gene mutations are sufficient to block nodulation. Therefore, loss of the syndrome could be quite frequent, and could be selectively favored under some environmental conditions. Additionally, the syndrome is quite complex, involving the production of a novel organ, the nodule, a novel organelle, the ''symbiosome'' (consisting of a primarily host-derived peribacteroid membrane surrounding nitrogen-fixing bacteroids; Verma, Hu, and Zhang, 1992), and nodule-associated proteins (termed ''nodulins''; Legocki and Verma, 1980), some of which may be the products of novel genes. That such a complex syndrome could arise multiple times might seem implausible; however, nitrogen-fixing symbioses almost certainly have arisen independently elsewhere in the dicots, in genera of Betulaceae, Casuarinaceae, Coriariaceae, Datiscaceae, Elaeagnaceae, Myricaceae, Rhamnaceae, Rosaceae, and Gunneraceae (Sprent and Sprent, 1990; Baker and Mullin, 1992). Although recent phylogenetic studies suggest that all of these families, with the exception of Gunneraceae, are more closely related to each other and to Leguminosae than was once thought (Soltis et al., 1995), the vast majority of species in the clade that includes these families do not nodulate, making it likely that multiple origins have occurred. A hypothesis of a separate origin of nodulation in Leguminosae from other families is also supported by nodule anatomy (Sprent, 1989; Sprent and Sprent, 1990). Also bolstering the plausibility of independent origins of nodulation within the family is the fact that nodulation is only one of several strategies (others include mycorrhizal assocations) used by legumes to satisfy nitrogen requirements that are very high throughout Leguminosae relative to other plant families (McKey, 1994).

Nodule morphology is known to be under control of the host plant (Sprent and Sprent, 1990), and thus could provide useful characters for assessing both taxonomic relationships and nodule homologies. Corby (1981, 1988) has classified the diverse types of nodules found in the family, and these can be mapped onto the *rbc*L tree, though the task is complicated by the ability of some tribes, genera, or even species to form more than one type of nodule, and by absence of data for particular species sampled here. The type of nodule found in Mimosoideae and associated Caesalpinioideae (e.g., *Erythrophleum*) is Corby's (1988) ''caesalpinioid'' type. This nodule type is also found in *Chamaecrista* and is the plesiomorphic nodule type among Papilionoideae. Thus at the level of gross nodule structure, nodules appear to be homologous throughout the family. Further study of nodule homology is warranted, such as the application of tests using orthology relationships and expression patterns of gene families encoding nodulin proteins (Doyle, 1994).

While the caesalpinioid nodule is the only common type among Caesalpinoideae and Mimosoideae, several other types occur in Papilionoideae, and the distribution of these classes correlates reasonably well with the topology of the *rbc*L tree. The caesalpinioid nodule has an apical meristem and is indeterminate in its growth; other indeterminate types are the highly branched crotalarioid nodule common both in the Genistoid Alliance and in the TH tribes, and the unique lupinoid nodule of *Lupinus.* Determinate types are round, nonbranching nodules; these include the relatively large desmodioid type, the very robust mucunoid nodule, and the small, clustered aeschynomenoid type. The aeschynomenoid nodule is found in Aeschynomeneae and Adesmieae, but is also reported from *Dalbergia* (Dalbergieae), which forms a clade with representatives of these tribes in the *rbc*L tree. This nodule type is not reported, however, from the other member of Dalbergieae in this clade, *Machaerium* (desmodioid nodules), nor from the sister to this clade, Amorpheae (caesalpinioid nodules). Desmodioid (and mucunoid) nodules are typical of the Millettieae/Phaseoleae clade, with approximately the same distribution as the pseudoracemose inflorescence (nodule type is unknown for *Austrosteenisia*, the only member of this clade that may lack pseudoracemes). These nodules occur sporadically elsewhere, however, for example in Loteae and in some *Machaerium* species (Dalbergieae). Desmodieae and Phaseoleae differ from most other legumes thus far studied in transporting nitrogen as ureide compounds rather than as amides (Verma and Stanley, 1989). Relatively few taxa have been studied for this trait, and the topology of the *rbc*L tree suggests other groups, such as Indigofereae and Psoraleeae, that warrant study for this biochemical difference.

Agreement with other hypotheses—Many aspects of the *rbc*L trees agree with topologies from the largely nonmolecular preliminary cladistic analysis of Chappill (1995; Fig. 3). Agreement with that analysis was better than with the less comprehensive analysis of Tucker and Douglas (1994; Fig. 4), which emphasized floral ontogenetic characters. It should be clear from the above discussion that the topology of the *rbc*L tree contains features that are in substantial agreement with many current hypotheses of legume relationships. This extends the general conclusion that has been made previously, at levels of individual tribes (e.g., Phaseoleae: Doyle and Doyle, 1993) or groups of tribes (e.g., Lavin, Doyle, and Palmer, 1990) that previous work on the family, starting with that of Bentham (1865), has provided a solid foundation for molecular and other phylogenetic studies. The strength of this foundation can only partly be measured by the nat-

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uralness (or lack thereof) of tribal and subfamilial taxonomies; understanding of evolutionary relationships by Bentham and later workers was often explicitly subordinated to the need for a workable classification system. The cladistic method provides the tools for building on that foundation, and, in particular, for assessing relationships among distantly related groups, especially when data can be obtained from characters whose homologies can be assessed with confidence over long evolutionary periods. The chloroplast genome has provided such data, but the weak support for many clades in the *rbc*L tree suggests that data from additional genes will be required to construct rigorous hypotheses for some groups. Data from phylogenetically independent nuclear genes will be of particular interest.

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